Blepharitis

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SOAP Format

- S = Subjective
  - AKA symptoms
  - Everything the patient says
- O = Objective
  - AKA signs
  - All the doctors observes and all the test results
- A = Assessment
  - AKA diagnosis (or diagnoses)
  - What is wrong with the patient
- P = Plan
  - For each diagnosis, what is going to be done to help the patient

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Blepharitis

- Anterior blepharitis - inflammation centered around eyelashes and follicles
  - Staphylococcal
  - Seborrheic
  - Mixed
- Posterior blepharitis - inflammation involving meibomian glands and gland orifices
  - Meibomian seborrheic
  - Meibomian gland dysfunction (AKA meibomianitis, meibomitis)
- Angular blepharitis
  - Staphylococcal
  - Moraxella

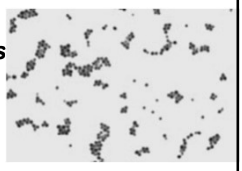
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Staphylococcal Blepharitis

Etiology

Infection of the lash follicles and lid margins

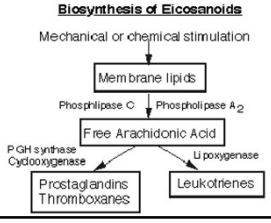
- **With *Staphylococcus aureus* and/or**
  - ***S. aureus* is normally present on lid margins of small number of individuals**
- **overgrowth of *Staphylococcus epidermidis***
  - ***S. epidermidis* normally present on lid margins (normal flora) of most individuals**
- **→ inflammatory response by the body**



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Staphylococcal Blepharitis

- **Secondary mechanism of inflammation**
  - **Production by the bacteria of lipase**
  - **→ breaks down the lipids in the tears**
  - **→ free fatty acids (FFAs)**
  - **→ trigger inflammatory and irritative response**



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Staphylococcal Blepharitis

Inflammatory response:

- **Dilation of blood vessels (redness)**
- **Increased vascular permeability (edema + blood proteins)**
- **Movement of PMNs from blood into tissue**
- **If inflammatory response is caused by bacteria, lots of PMNs move from blood into tissues to fight infection through phagocytosis**
- **→ dead PMNs + protein clot = pus**



- ****** Presence of pus = bacterial infection**

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PMNs = polymorphonucleocytes (white blood cells)

Staphylococcal Blepharitis

Subjective

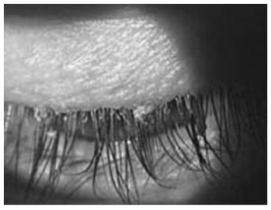
- Depends on severity
- Chronic irritation and burning
- FB (foreign body sensation)
- Tender lids (may have recurrent hordeola or chalazia)
- Patient may notice crusts on lashes
- The lids may be stuck together in morning
- Patient may complain of red, irritated lids

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Staphylococcal Blepharitis

Objective

- Crusts along the lash line at the base of the lashes
- Dried pus

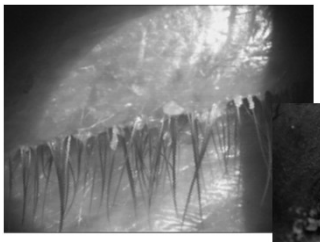


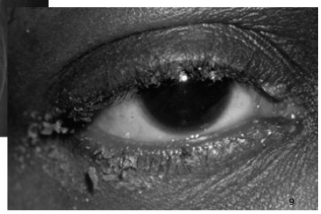


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Staphylococcal Blepharitis

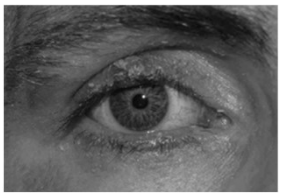
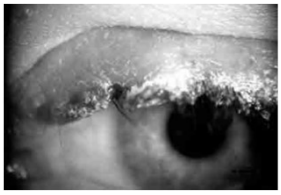
More crusts





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Staphylococcal Blepharitis

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Staphylococcal Blepharitis

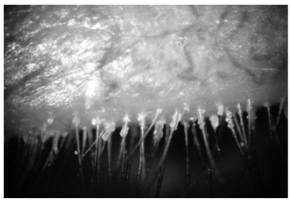
- Crust can encircle lash = collarette



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Staphylococcal Blepharitis

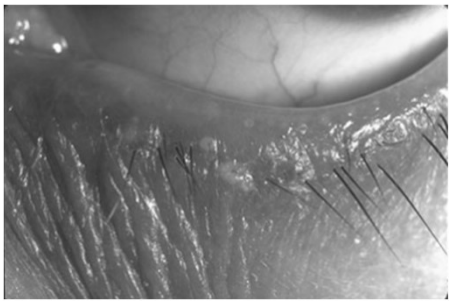
- Crust at the lash base – can also appear 'oily'




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Staphylococcal Blepharitis

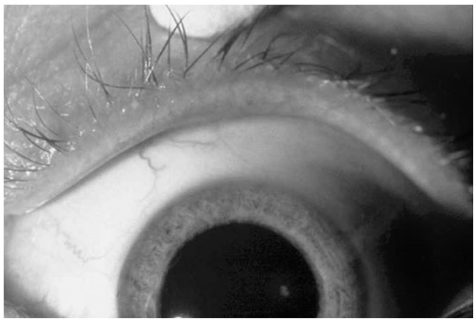
- Infected follicles and madarosis (loss of lashes)



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Staphylococcal Blepharitis

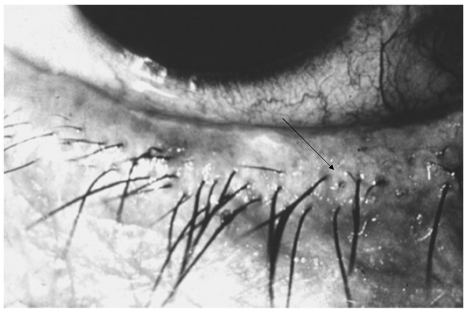
- Infected follicles



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Staphylococcal Blepharitis

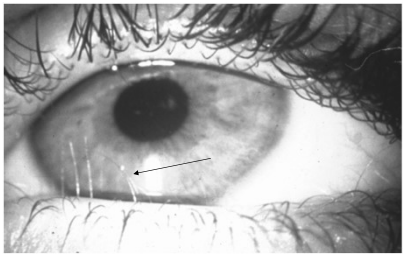
- Ulcers at the base of the lashes



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Staphylococcal Blepharitis

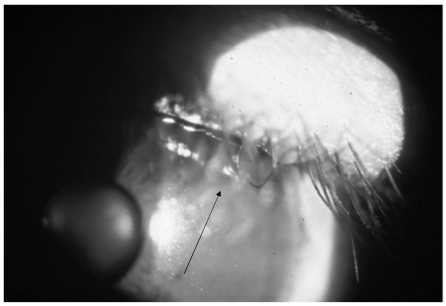
- Damage to the lash follicles → trichiasis (lashes that point towards the cornea and may actually scratch the cornea)



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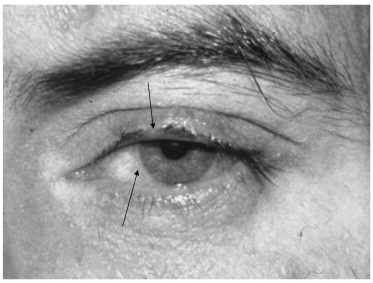
Staphylococcal Blepharitis

- Trichiasis - lashes need to be epilated if touching cornea



Staphylococcal Blepharitis

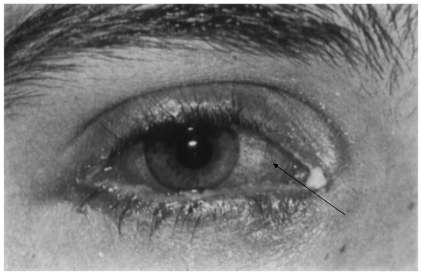
- Red and thickened (tylosis) lid margins
  - Seen mostly with chronic disease
  - Also note madarosis and conjunctivitis
- Lid margins may become indurated (hardened) over time



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Staphylococcal Blepharitis

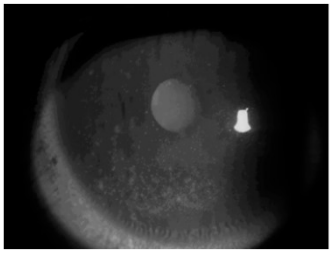
- May have secondary conjunctivitis (usually papillary)
  - Sometimes called staphylococcal blepharoconjunctivitis



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Staphylococcal Blepharitis

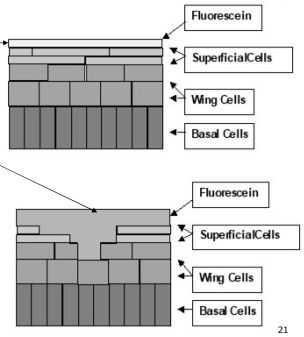
- Punctate epithelial erosions (PEE)
  - Result from reaction to bacterial exotoxins
- PEE in inferior third of the cornea
- PEE stains with fluorescein (green color with colbalt blue filter)



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Staphylococcal Blepharitis

- Normal
- PEE (punctate epithelial erosions)
- (AKA SPK - superficial punctate keratopathy)
- Divits in the corneal epithelium that cause fluorescein to fluress



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Staphylococcal Blepharitis

- Diagnosis
  - Usually based on clinical findings
  - Severity will dictate appropriate treatment

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Staphylococcal Blepharitis

Treatment - lid hygiene

- Remove crusts
- Remove bacteria
- Increase blood circulation to lid margin
- Two step process - warm compresses and lid scrubs



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Staphylococcal Blepharitis

Warm compresses

- Soak lids with warm, wet compress (eyelids closed)
- 10-15 min bid-qid



Lid scrubs

- Diluted baby shampoo and water or
- Commercially available cleansers
- Scrub lid margins
- Bid to qid after warm compresses
- Hypochlorous acid 0.02%



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Staphylococcal Blepharitis

Baby Shampoo – the 'old' treatment modality

- **The baby shampoo is first diluted one-to-one with water in 'cup' in palm of hand.**
- **Then mixed by rubbing with clean fingertips and applied in gentle oval scrubbing motion to margin and eyelash base of closed eyelid for 1 minute, followed by fresh water facial rinse.**
- **Baby shampoo can alternatively be diluted in container (e.g. plastic cup) and scrubbing performed using washcloth wrapped around finger (after dipping it in the diluted shampoo). A cotton tip applicator may be used alternatively.**
- **NOT recommended anymore due to drying effects**



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Hypochlorous acid 0.02%

- **Ocusoft HypoChlor™**
- **Spray or gel**
- **Used as needed or daily**
- **Can be used with lid wipes**
- **Spray form often used AM**
- **Gel form often used PM**
- **Generic versions as well**



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Staphylococcal Blepharitis

Antibiotic ointment applied to lid margins

- **Bacitracin**
  - **Cell wall inhibitor**
- **Erythromycin**
  - **Macrolide**
- **Gentamicin**
  - **Aminoglycoside**
- **Tobramycin**
  - **Aminoglycoside**



Regimen

- **Moderate: qhs or bid**
- **Severe: tid or qid**

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Prescription Writing

Christina Gosses, OD
354 NW Orange Lane
Chicago, IL 60631

Name_Sherman Pierce____ Age_22_ Date_09/22/24_
Address_2043 College Way, Forest Grove OR 97116__

℞ Bacitracin ophthalmic ung
Dispense: one 3.5 gram tube
Instructions: apply 1/4 inch to each lid margin of both eyes twice a day for 10 days

X No Refill _____,O.D._____,O.D.
Refill x ___ Dispense as Written Product Selection Permitted

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Staphylococcal Blepharitis

Lubricants
- Dry eye secondary to lid condition
- Artificial tears qid or prn

Oral Antibiotics
- Valuable when unusually severe
- Doxycycline: 100mg PO bid x 7-10 days
  - Tetracyclines inhibit lipase synthesis by bacteria
  - Doycycline inhibits Staphylococcal exotoxin-induced cytokines and chemokines
  - Are helpful even if bacteria are resistant to antibiotic effect

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Staphylococcal Blepharitis

Follow-up
- 1-3 weeks depending on severity and medications prescribed
- Consider follow-up 2-3 weeks later
- If cornea involved, follow-up reduced to days

Patient education
- Chronic condition - Usually exacerbations and remissions
- Goal: to control not cure
- Importance of daily lid hygiene
- Medication for acute flare-ups

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BlephEx – Blepharitis treatment option

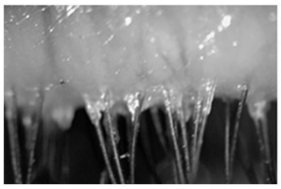
- Rysurg
- Medical grade disposable micro-sponge spins along edge of eyelids and lashes
- Removes scurf/debris and exfoliates eyelids
- Proparacaine usually instilled prior to treatment
- Eyes rinsed afterwards
- 6 -8 minute procedure

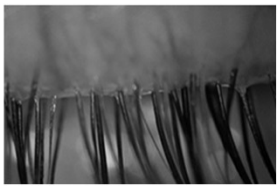


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Blephex

Before **After**





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Staphylococcal Blepharitis Associated Marginal Infiltrate

- White blood cells infiltrate cornea between epithelium and stroma (subepithelial)
  - White spot in corneal periphery with clear area between it and limbus
  - Does not stain with fluorescein
- When the white cells die, they release enzymes that can damage the epithelium above the infiltrate
  - The infiltrate may then stain in a sporadic punctate pattern



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Staphylococcal Blepharitis Associated Marginal Infiltrate

Hypersensitivity to staphylococcal toxins

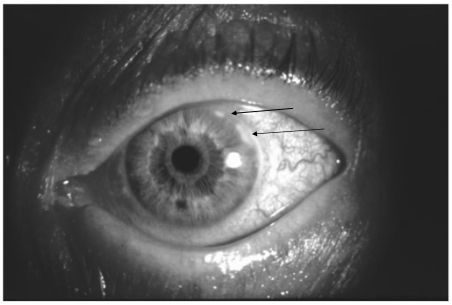
Ag-Ab-Complement triggered

- **Ag (staph product) +**
- **Ab (produced in conjunctiva) +**
- **complement components (from limbal blood vessels) →**
- **C3a and C5a (chemotactic factors for white cells from the limbal blood vessels) →**
- **White spot in cornea of an accumulation of white cells**

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Staphylococcal Blepharitis Associated Marginal Infiltrate

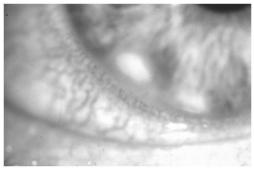
- **Marginal infiltrates**

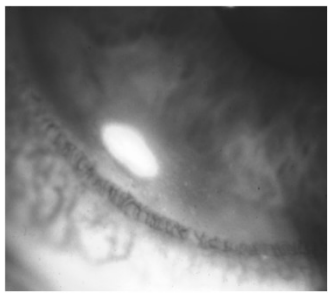


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Staphylococcal Blepharitis Associated Marginal Infiltrate

A marginal infiltrate that does not stain





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Staphylococcal Blepharitis Associated Marginal Infiltrate

Treatment

- **Treat the staphylococcal blepharitis and add:**
- **Topical steroid q2h to qid**
  - **Prednisolone acetate**
  - **Dexamethasone**
  - **Loteprednol**



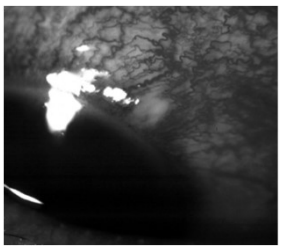
Or…

- **Topical antibiotic/steroid combination such as tobramycin with dexamethasone (Tobradex), or tobramycin with loteprednol (Zylet) q2h to qid**

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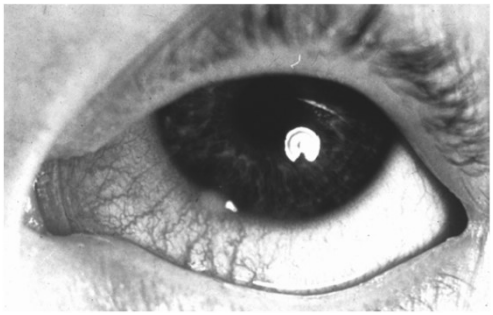
Staphylococcal Blepharitis Associated Phlyctenule

- **Hypersensitivity reaction to staph toxins**
- **Type IV reaction**
- **Raised white lesion on conjunctiva, limbus, or cornea**
- **Can move over cornea, pulling blood vessels and scarring as it goes**



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Staphylococcal Blepharitis Associated Phlyctenule



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Staph Blepharitis Associated Phlyctenule

Treatment

- Treat the staphylococcal blepharitis and add:



- Topical steroid q2h to qid
  - Prednisolone acetate
  - Dexamethasone
  - Loteprednol

Or…

- Topical antibiotic/steroid combination such as tobramycin with dexamethasone, or tobramycin with loteprednol q2h to qid

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Staph Blepharitis Associated Phlyctenule

- Note – several years ago a primary cause of phlyctenule was tuberculosis (TB)
- Patients who were:
  - Recent immigrants from areas endemic with TB, or
  - Individuals who acquired active TB locally
- Now, Staphyloccus has once again become a primary cause of phlyctenule

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Seborrheic Blepharitis

Etiology

- Unknown
- Often associated with seborrheic dermatitis
  - Chronic disorder of the skin/head areas where sebaceous glands are abundant
  - Dandruff (AKA scurf) - excessive shedding of dead skin cells of the scalp - is often a symptom of seborrheic dermatitis



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Seborrheic Blepharitis

Subjective

- **Frequently asymptomatic**
- **Possible morning "mattering"**
- **Dry eye complaints (poor tear film)**
- **Symptoms may depend on other concurrent processes**

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Seborrheic Blepharitis

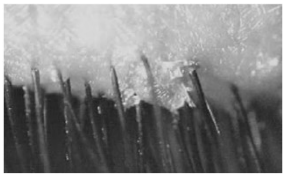
Objective

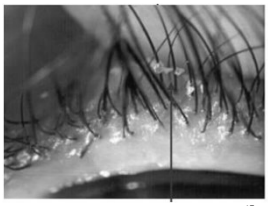
- **Greasy, waxy scales at the base of the lashes that flake off easily (dandruff-like)(AKA scurf)**
- **Skin of the lid a little greasy**



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Seborrheic Blepharitis

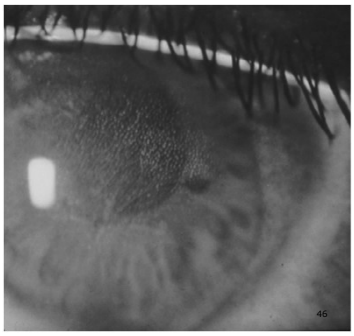




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Seborrheic Blepharitis

- Interpalpebral PEE

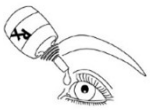


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Seborrheic Blepharitis

Treatment:

- Lid hygiene
- Lubricants
- Follow-up 2-4 weeks
- Patient education on chronicity
  - Lid hygiene daily





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Mixed Blepharitis

- Seborrheic blepharitis with associated staphylococcal superinfection
- Plan
  - Same as for staph blepharitis

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Meibomian Seborrheic Blepharitis

Etiology
- Unknown
- Excessive oil production
- Quality of oil is abnormal

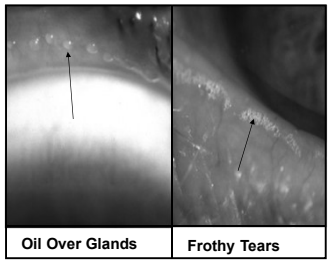
Subjective
- Dry eye complaints

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Meibomian Seborrheic Blepharitis

Objective
- Frothing of tears
- "Oil slick" in tear film
- TBUT may be reduced
- Conjunctival injection
- Lid tissue appears normal

Treatment
- Keep lid margins clean
- Lubricants

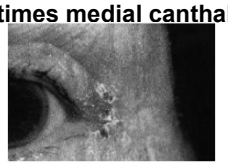


Oil Over Glands | Frothy Tears

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Angular Blepharitis

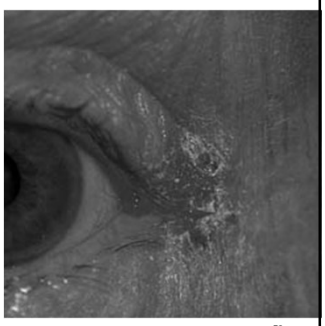
- Localized eczematoid inflammation of lid at outer canthus and sometimes medial canthal region
- Staphylococcal form:
  - Dry and scaly
- Moraxella form:
  - Wet
  - Macerated lid with whitish, frothy discharge
- Angular blepharitis may involve conjunctiva
- All forms of angular blepharitis call for treatment with antibiotic ointment
  - May need to try different ABs based on bacteria type



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Angular Blepharitis

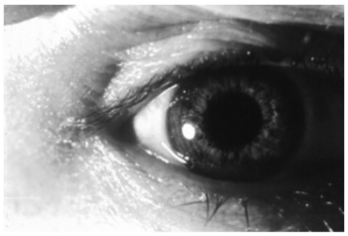
- Chronic lid irritation
- Skin drying, scaling, cracking in outer canthi
- Cause
  - *Moraxella lacunata*
  - *Staphylococcus epidermidis*
  - *Staphylococcus aureus*



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Angular Blepharitis

- *Moraxella*: zinc sulfate 0.25% solution
- *Staphylococcus*: bacitracin or erythromycin



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THANK YOU!

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